# Component Descriptions and Purpose

**1. ESP32-WROOM-32U Microcontroller**

Purpose: The ESP32 is a powerful microcontroller used for controlling various devices in the circuit. It acts as the main brain of the setup, managing communication with the motor drivers to control motor actions. The GPIO pins are used to send PWM signals to the BTS7960 motor drivers to adjust motor speed and direction.

Connections: The ESP32 is powered by the 3.3V output of one of the LM2596 buck converters and is connected to the motor drivers using labeled GPIO pins for control.

**2. BTS7960 Motor Drivers (4 units)**

Purpose: The BTS7960 is a high-power H-bridge motor driver module capable of handling the high current required to drive motors. It is responsible for receiving the control signals from the ESP32 and driving the 24V motors accordingly.

Connections: Each motor driver is connected to the corresponding GPIO pins on the ESP32 for control, as well as the power supply from the battery through the XH-M609 protection module. Each driver is connected to its own 24V motor for driving operations.

**3. 24V 22k RPM 200W Motors (4 units)**

Purpose: These high-speed motors are used to convert electrical energy into mechanical energy. They are driven by the BTS7960 motor drivers, providing movement or rotational force for the connected load.

Connections: Each motor is connected to one of the BTS7960 motor drivers via the motor terminals (M+ and M-). The motor driver is responsible for varying the voltage and direction supplied to the motor.

**4. LM2596 DC-DC Buck Converters (2 units)**

Purpose: These buck converters step down the 24V supply from the battery to lower voltages as

required by other components in the circuit. One converter steps down to 3.3V to power the ESP32, and the other provides 12V for accessories like sensors or external components.

Connections: The LM2596 converters are connected to the output of the XH-M609 battery discharge protection module. Their outputs are connected to provide power to the ESP32 (3.3V) and any future accessories (12V).

**5. 30A Resettable Circuit Breaker**

Purpose: This component is used for overcurrent protection in the circuit. It helps prevent damage by disconnecting power when the current exceeds 30A, which could otherwise damage components.

Connections: The circuit breaker is connected between the main battery and the XH-M609 discharge protection module to ensure that the entire circuit is protected.

**6. XH-M609 Battery Discharge Protection Module**

Purpose: The XH-M609 module is used to protect the battery from deep discharge, which can be harmful to its lifespan. It monitors the battery voltage and disconnects the load if the voltage falls below a certain threshold.

Connections: It is connected between the 30A circuit breaker and the rest of the components in the circuit, including the LM2596 converters and BTS7960 motor drivers. Its output ensures that the connected devices receive stable power without discharging the battery excessively.

**7. Battery (24V)**

Purpose: The battery is the main power source for the entire circuit, providing the necessary voltage and current to all components.

Connections: The battery is connected to the 30A circuit breaker, and then through the XH-M609 battery discharge protection module to the various other components, including the LM2596 buck converters and BTS7960 motor drivers.